

Julia Hughes

📞 (626)-491-9105 ✉️ jh930130@gmail.com 🔗 [jmhughes-eng.github.io](https://github.com/jmhughes-eng) 🇺🇸 U.S. Citizen

Objective

Mechanical engineering student with industry internship experience and a strong hands-on foundation in CAD, mechanical design, engineering drawings, and manufacturing. Seeking an aerospace engineering internship where I can make an impact on challenging, mission-driven projects.

Education

California State Polytechnic University, Pomona

August 2024 – May 2028

Bachelor of Science in Mechanical Engineering - GPA: 3.66

Pomona, California

Related Coursework: Vector Dynamics, Vector Statics, Mechanics of Materials, Fluid Dynamics, Thermodynamics, Engineering Design, Manufacturing, Engineering Graphics & Visualization, Computer-Aided Computations

Projects & Experience

Glenair – Product Engineering Intern

May 2026 – August 2026

- Supported the “Mighty Mouse” Product Engineering team in the design and production of micro miniature electrical connectors by creating bills of materials, drawing change notices, and customer CAD models.
- Created connector models and revised sales drawings using SolidWorks and CADKEY while collaborating with manufacturing and quality engineers to resolve production issues and investigate customer-reported quality issues.
- Produced official sales drawings and bills of materials for a developmental size 24 bayonet-coupling connector and coordinated the production of engineering samples for a size 12 version with manufacturing engineers.

Cal Poly Rose Float – Mechanism Lead

September 2024 – Present

- Designed a four-bar linkage to power a diving-goose animation for the 2027 Rose Parade and presented two preliminary design reviews to program alumni and professional engineers.
- Led a team with varied technical experience through mechanism fabrication, coordinating tasks to meet deadlines.
- Trained new and returning team members in manufacturing, lab safety, and fabrication processes, including welding, milling, and power-tool operation.

Project Lazarus – Payload Team

September 2025 – Present

- Contributed to a 7-member team to develop and integrate a 1.0 kg deployable drone payload for the FAR Unlimited rocketry competition.
- Led the research, design, and integration of the drone’s folding and deployment mechanism (“airbrakes”) for compact storage and reliable deployment from the rocket.
- Built an FDM 3D-printed working prototype and contributed to the structural and electronic integration of the drone in the rocket using SolidWorks and OpenRocket.
- Developed signal-range and vibration-resistance test procedures; the payload’s successful deployment contributed to a third-place overall finish at FAR Unlimited.

University Advancement Office – Student Assistant, Special Events

February 2026 – Present

- Organized commencement data for 7,500 graduating students in Excel to support commencement coordination.
- Assisted Special Events staff and student workers with event operations and on-site technical support.

Skills

CAD/Design: SolidWorks, Fusion 360, Onshape, CADKEY, GD&T

Analysis/Programming: SolidWorks Simulation, OpenRocket MATLAB, Arduino, Excel VBA, JavaScript

Manufacturing: Metal Fabrication (SMAW Welding, Machining), FDM 3D Printing, Soldering

Honors & Awards

Yee Award – Recognized for Excellence in “Hands-On” Projects by the Aerospace Engineering Department

May 2026

President’s List – Awarded for a GPA of 3.5+ across two consecutive semesters

May 2025

Memberships & Affiliations

American Society of Mechanical Engineers – Membership Chair, Chapter Member

May 2025 – Present

Cal Poly Pomona Kings Chess Club – Scheduler

May 2025 – Present